B. TECH-ME-III-I Sem

23A03504d	WORK STUDY AND ERGONOMICS (Professional Elective-I).	L	T	P	C
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Course objectives: The objectives of the course are	
1	To develop concepts related to principles of productivity & work study as a tool for increasing the efficiency and effectiveness in organizational systems.
2	To study the existing method, compare and propose a new method.
3	To provide the usage of the various tools and techniques used in work measurement.
4	To develop basic ideas of ergonomics and its design.
5	To develop concepts related Man-Machine Interfaces and Design of Displays and controls.

COURSE OUTCOMES: Upon completion of this course, students should be able to:		
1	Recollect the basic concepts of productivity, work content and work study and define the objective and scope of Work Study.	L1, L2
2	Define the various charts and to construct the charts on the basis of present method and develop a new / proposed method and identify the unnecessary movements.	L1, L3 L4
3	Explain the basic work measurement techniques and to gain knowledge of measurement of work, rating and imbibe the concept of allowance in estimating Standard Time	L1, L2
4	Determine the basic concepts of Ergonomics and demonstrate a sound knowledge of Ergonomics in engineering applications.	L2, L3
5	Demonstrate a sound knowledge of Man-Machine Interfaces and design of displays and controls in engineering systems	L3, L4

UNIT – I

Productivity and Work Study: Definition of productivity, task of management, productivity of materials, land, building, machine and power, factors affecting the productivity, work content, basic work content, excess work content, how manufacturing job is made up, work content due to excess product and process, ineffective time due to short comings on part of the management. Definition, Objective and scope of Work Study: Work study and management, work study and work.

UNIT – II

Method Study: Definition, objective and scope of method study, activity recording and tools, Recording tools: Out Line Process Chart, Flow Process Chart, Flow diagram, String Diagram, Travel Chart, Multiple Activity Chart, Two- Handed process chart. Principles of Motion Economy: Introduction, Classification of movements. Two- hand process chart, Micromotion study, Therbligs, SIMO Chart. Special Charts: Cyclegraph and Chronocycle graph - development, definition and installation of the improved method. Work Measurement: Definition, objectives, work measurement techniques.

Work sampling – Need, confidence levels, and sample size determination, conducting study with problems

UNIT – III

Time study - Definition, time study equipment, selection of job, steps in time study. Breaking jobs into elements, recording information. Rating: Systems of rating, standard rating, standard performance, scales of rating. Allowances: Standard time determination, predetermined motion time study (PMTS), factors affecting rate of working, problems on allowances.

UNIT – IV

Introduction to Ergonomics: Human factors and ergonomics, psychology, engineering, bio mechanics, industrial design, graphics design, statistics, operation research and anthropometry Morphology of design and its relationship with cognitive abilities of human being. Physical Ergonomics : human anatomy, and some of the anthropometric, physiological and bio mechanical characteristics as they relate to physical activity. Cognitive: mental processes, such as perception, memory, reasoning, and motor response, mental workload, and decision-making. Organizational ergonomics: optimization of socio-technical systems, including their organizational structures, policies, processes. Communication, work design, design of working times, teamwork, cooperative work, and new work programs. Environmental ergonomics: human interaction with the environment-characterized by climate, temperature, pressure, vibration, light.

UNIT – V

Man-Machine Interaction; Man-Machine interaction cycle, Man-machine interfaces, Displays factors that control choice of display, visual displays- qualitative displays; moving pointer displays, moving scale displays, digital displays Indicators, auditory displays, tactile displays. Factors affecting effectiveness of displays. Quantitative displays, check- reading displays, representational displays. Types of controls and their integration with displays. Design guidelines for displays and controls: viewing distance, Illumination, angle of view, reach etc., general design checklist for displays and controls. Standards for ergonomics in engineering and design, displays and controls.

TEXT BOOKS

1. Introduction to Work Study – ILO, 4th edition 1992

2. Mark. S. Sanders and Ernest. J McCornick. “Human Factor in Engineering and Design”, McGraw-Hill Book Co., Inc., New York, 1993

REFERENCE BOOKS

1. S. Dalela and Sourabh, “Work Study and Ergonomics”. Standard publishers 2013

2. Wesley Woodson, Peggy Tillman and Barry Tillman, “Human Factors Design Handbook”, McGraw-Hill; 2ndedition, 1992

3. Ralph M. Barnes, “Motion and Time Study”, Wiley International, 7th Edition.

4. Mark S. Sanders and Ernest J. McCormick , “Human Factors in Engineering Design” 4th edition, 2013.

5. B. Niebel and Freivalds, Niebel’s Methods Standards and Work Design, McGraw-Hill, 12th Edition, 2009,

Online Learning Resources:

https://youtu.be/b05FPBjFH6A?si=dWB1YOLOmSMRBSX7

https://youtube.com/playlist?list=PLLy_2iUCG87BbIF6sF5sy_ZZLFoUcnncb&si=n1NAnFTtiocc9vtK

https://youtube.com/playlist?list=PLuF8VVHesRxXBZzQpQSzvJI7eM_SduxwR&si=j2vyTNYybgvXrDiy